

Maggie Makar

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Education

Massachusetts Institute of Technology Ph.D, Electrical Engineering and Computer Science	Expected 05/2021
Massachusetts Institute of Technology Master of Science, Electrical Engineering and Computer Science Thesis title: <i>Learning the probability of activation in the presence of latent spreaders</i>	05/2017
University of Massachusetts, Amherst B.S., Mathematics and Economics, <i>Summa Cum Laude</i>	02/2013

Employment

Research Intern , Microsoft Research	06/2018 - 09/2018
Research Assistant , Brigham and Women's Hospital	02/2013 - 08/2015
Research Assistant , Political Economy Research Institute	09/2012 - 12/2012
Research Intern , Donahue Institute,	08/2012 - 09/2012
Research Assistant , Center for Intelligent Information Retrieval	02/2012 - 08/2012

Publications

* denotes equal contribution.

1. **M. Makar**, A. Swaminathan, E. Kiciman, "A Distillation Approach to Data Efficient Individual Treatment Effect Estimation". *AAAI-19*
2. **M. Makar**, J. Guttag, J. Wiens, "Learning the Probability of Activation in the Presence of Latent Spreaders". *AAAI-18, Oral presentation*
3. **M. Makar***, J. Antonelli*, Q. Di, J. Schwartz, D. Cutler, F. Dominici, "Estimating the Causal Effect of Fine Particulate Matter Levels on Death and Hospitalization: Are Levels Below the Safety Standards Harmful?". *Epidemiology*. 2017 Sep 28(5): 627-634
4. J. Oh*, **M. Makar***, et. al, "A generalizable, data-driven approach to predict daily risk of Clostridium difficile infection at two large academic health centers". *Infection Control and Hospital Epidemiology*. Apr 2018; 39(4), 425-433.
5. Z. Obermeyer, **M. Makar**, S. Abujaber, F. Dominici, S. Block, D. Cutler, "Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer". *Journal of the American Medical Association*. 2014 Nov 12; 312(18): 1888-1896
6. **M. Makar**, M. Ghassemi, D. Cutler, Z. Obermeyer, "Short-term mortality prediction for elderly patients using Medicare claims data". *International Journal of Machine Learning and Computing*. 2015 Vol. 5(3): 192-197

7. B. Powers, **M. Makar**, D. Cutler, Z. Obermeyer, “Cost Savings Associated with Expanded Hospice Use Among Medicare Beneficiaries with Poor Prognosis Cancer”. *Journal of Palliative Medicine* 2015 May; 18(5):400-1
8. Z. Obermeyer, S. Abujaber, **M. Makar**, S. Stoll, S. Kayden, L. Wallis, T. Reynolds, “Emergency care delivery in 59 low- and middle-income countries: Systematic review”. *WHO Bulletin*. May 2015
9. Z. Obermeyer, B. Powers, **M. Makar**, N. Keating, D. Cutler, “Physician revealed preferences for hospice are strongly related to their patients’ enrollment in hospice”. *Health Affairs*. 2015 Jun 1; 34(6):993-1000
10. Z. Obermeyer, A. Clarke, **M. Makar**, J. Schuur, D. Cutler, “Emergency Care Use and the Medicare Hospice Benefit for Individuals with Cancer with a Poor Prognosis”. *Journal of the American Geriatrics Society* 2016 Feb; 64(2):323-9
11. S. Skamene, I. Agarwal, **M. Makar**, et al., “Impact of a dedicated palliative radiation oncology service on the use of single fraction and hypofractionated radiation therapy among patients with bone metastases”. *Annals of Palliative Medicine*, 2017

Workshops and poster sessions

1. *Learning the influence of individual Clostridioides Difficile infections*. Accepted for presentation at the Infectious Diseases week, 2019.
2. *Utility maximizing bounds on potential outcome estimation*. Microsoft AI breakthroughs workshop, 2019.
3. *A distillation approach to data-efficient individual treatment effect estimation*. Machine learning for health workshop, NeurIPS, 2018.
4. *A distillation approach to data-efficient individual treatment effect estimation* Women in Machine Learning workshop, NeurIPS, 2018.
5. *Learning the probability of activation in the presence of latent spreaders*. Women in Machine Learning workshop, NeurIPS, 2017.
6. *A data-driven approach to predict daily risk of Clostridium difficile infection at two large academic healthcare centers*. Infectious disease week, 2017
7. *Learning the probability of activation in the presence of latent spreaders*. Statistics and Data Science conference at MIT, 2017
8. *Estimating the Causal Effect of Lowering Particulate Matter Levels Below the National Ambient Air Quality Standards on Health Outcomes*. Joint Statistical Meeting (JSM), 2016
9. *Impact of a Dedicated Palliative Radiation Oncology Service on the Use of Single-Fraction and Hypofractionated Radiation Therapy Among Patients With Bone Metastases*. ASTRO, San Antonio, November 2015
10. *Learning meaningful representations from medical claims data*. Machine learning for healthcare workshop, NeurIPS, 2015
11. *Short-term mortality prediction for elderly patients using Medicare claims data*. Max Planck Institute, 2015

Invited talks

1. *A distillation approach to data-efficient individual treatment effect estimation*. Spotlight talk at AAAI, Hawaii, February 2019.

2. *A distillation approach to data-efficient individual treatment effect estimation*. Microsoft Research, Redmond, August 2018.
3. *Spread of Contagions in the Presence of Latent Spreaders: Identifying Hidden Culprits and Learning the Probability of Infection*. University of Massachusetts, Amherst, September 2018
4. *Learning the probability of activation in the presence of latent spreaders*. Spotlight talk at AAAI, Hawaii, February 2018.
5. *Spread of Contagions in the Presence of Latent Spreaders*. Computational Cardiovascular Research Group, MIT, October 2017.
6. *Inference on graphs with missing nodes*. University of Michigan in Ann Arbor, July 2016
7. *Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer*. Harvard School of Public Health. October 2014

Professional Service

TEACHING

Co-organizer, instructor: Whirlwind of Machine learning, MIT	IAP 2017
Teaching assistant: Machine learning for Healthcare, MIT	Spring 2017

MENTORSHIP

Advaith Anand, Masters of Engineering candidate	2018-2019
Emily Mu, Masters of Engineering candidate	2018-2019
Agni Kumar, Masters of Engineering Candidate	2019-2020
Meghana Kamineni, Undergraduate researcher	2019-2020

REVIEWING

International Conference on Machine Learning (ICML; *Top 5% reviewer award*)
 Neural Information Processing Systems (NeurIPS; *Top 5% reviewer award*)
 AAAI
 Machine Learning for health NeurIPS workshop
 Women in Machine Learning NeurIPS workshop
 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
 Journal of Health Economics

Awards

Harvey Fellowship (\$48,000)
 Ward McCarthy scholarship
 School of Behavioral and Social Sciences scholarship
 Economics Department Writing Award
 Life Members Scholarship

Computer and Language Skills

Computing: Python, and R
Languages: English, Arabic, and French